

SHETH L.U.J & SIR M.V COLLEGE
SUBJECT : Data Visualization with Tableau

Aim: Data Preparation and Transformation using Tableau Prep

A: Connect and Load Data into Tableau Prep

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists three text files: 'Student_Assessment...', 'Student_Assessment...', and 'Course_Credit_Mast...'. The 'Tables' pane shows 'Course_Credit_Master'. The main workspace displays a flow diagram with three data sources connected to a 'View and clean data' step. The 'Data Preview' tab is active, showing a table with 5 fields: 'Subject_Code', 'Subject_Title', 'Credit_Hours', and 'Passing_Marks'. The table contains 6 rows of data.

	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
1	CS101	Data Structures	4	40
2	CS201	Algorithms	4	40
3	EC101	Digital Logic	3	40
4	EC201	Signals & Systems	4	40
5	IT101	Database Systems	4	40
6				

The screenshot shows the Tableau Prep Builder interface with a 'Clean 1' step added to the flow. The 'Data Preview' tab is active, showing a table with 9 fields: 'Student_ID', 'Student_Name', 'Department', 'Assessment...', 'Exam_Date', 'Subject_Code', and 'Marks_Obtai...'. The table contains 16 rows of data.

Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject_Code	Marks_Obtai...
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35
S102	aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55
S103	Ananya Singh	CS		2025 09 15	IT101	75
S104	Kavya Reddy	CSE		2025 11 20	ME101	95
S105	priya VERMA	ECE		2025/09/15		
S106	Rahul Nair	Electronics		2025/11/20		

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B: Clean and Profile Data

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Search

Tables

- Course_Credit_Master

Clean 1 9 fields 16 rows

Select All Filter Values...

Automatic Split Custom Split...

Search

Student_ID	Student	Department	Assessment	Exam_Date	Subject_Code	Marks_Obtained
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35
S102	Aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55
S103	Ananya Singh	CS		2025-09-15	IT101	75
S104	Kavya Reddy	CSE		2025-11-20	ME101	95
S105	Priya Verma	ECE		2025/09/15		
S106	Rahul Nair	Electronics		2025/11/20		

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Search

Tables

- Course_Credit_Master

Clean 2 9 fields 14 rows

Select All Filter Values... Identify Duplicate Rows Rename Field

Search

Changes (1)

- Trim Spaces Full_Name Spaces trimmed from all values

Std_ID	Full_Name	Dept	Exam_Type	Assessment	Course
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201
S103	Kavya Reddy	CSE			IT201
S104	Priya Verma	ECE			
S106	Rahul Nair	Electronics			
S107	Rohan Gupta	Information Tech			
S108	Sneha Patel	IT			

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C: Group and Standardize Values

The screenshot shows the Tableau Prep Builder interface. The workflow consists of two 'Clean' steps. The 'Clean 2' step is currently selected, showing a 'Group Values' operation on the 'Dept' field. The 'Changes (2)' pane on the left indicates that 4 values were replaced by 3 values. The 'Table' pane on the right shows the resulting data with columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, and Course_Code. The 'Dept' field is grouped into three categories: Computer Science, Electronics, and Information Tech.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201
S102	Ananya Singh	Electronics	Midterm	25/04/2026	EC201
S103	Kavya Reddy	Information Tech			IT201
S104	Priya Verma				
S106	Rahul Nair				
S107	Rohan Gupta				
S108	Sneha Patel				

D: Merge Mismatched Fields

The screenshot shows the Tableau Prep Builder interface. The workflow consists of two 'Clean' steps followed by a 'Union 1' step. The 'Union 1' step is currently selected, showing a 'Union Results' operation. The 'Inputs' pane on the left shows the two sources being merged: 'Clean 2' and 'Clean 1'. The 'Union Results' pane on the right shows the merged data with columns: Table Names, Enrollment_Status, Exam_Date, Max_Marks, and Exam_Type. The 'Mismatching Fields' section indicates that there are 0 mismatching fields from 10 resulting fields.

Student_Name	Table Names	Enrollment_Status	Exam_Date	Max_Marks	Exam_Type	Course_Code	Score
AARAV SHARMA	Student_Assessments_Term1.csv	Active	15/09/2025	100	Midterm	CS101	85
AARAV SHARMA	Student_Assessments_Term1.csv	Active	20/11/2025	100	Final	CS101	92
PRIYA VERMA	Student_Assessments_Term1.csv	Active	15/09/2025	100	Midterm	EC101	70

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Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables

- Student_Assessment...

Flow Diagram:

```
graph LR; A[Student_Asses...] --> B[Clean 1]; B --> C[Union 1]; C --> D[Clean 5]; D --> E[Clean 6];
```

Clean 6: 10 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Rename Fields... Create Calculated Field...

Changes (0)

Student_Name	Table Names	Enrollment...	Exam_Date	Max_Marks	Exam_Type	Course_Code
Aarav Sharma	Student_Assessments_Term1	Active	15/09/2025	100	Final	CS101
AARAV SHARMA	Student_Assessments_Term2	Dropped	20/11/2025		Midterm	CS201
ADITYA RAO			10/02/2026			EC101
Ananya Singh			25/04/2026			EC201
Ananya Singh						IT101
Kavya Reddy						IT201
KAVYA REDDY						ME101
Priya Verma						
PRIYA VERMA						
Rahul Nair						
RAHUL NAIR						
Rohan Gupta						

Student_Name	Table Names	Enrollment_Status	Exam_Date	Max_Marks	Exam_Type	Course_Code	Score	Dept	Std_ID
AARAV SHARMA	Student_Assessments_Term1.csv	Active	15/09/2025	100	Midterm	CS101	85	Computer Science	S101
AARAV SHARMA	Student_Assessments_Term1.csv	Active	20/11/2025	100	Final	CS101	92	Computer Science	S101
PRIYA VERMA	Student_Assessments_Term1.csv	Active	15/09/2025	100	Midterm	EC101	70	Electronics	S102
PRIYA VERMA	Student_Assessments_Term1.csv	Active	20/11/2025	100	Final	EC101	78	Electronics	S102
ROHAN GUPTA	Student_Assessments_Term1.csv	Active	15/09/2025	100	Midterm	CS101	45	Computer Science	S103

E: Perform Aggregations in Prep

Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables

- Student_Assessment...

Flow Diagram:

```
graph LR; A[Student_Asses...] --> B[Clean 1]; B --> C[Union 1]; C --> D[Clean 6]; D --> E[Aggregate 1];
```

Aggregate 1: 0 fields

Filter Values...

Settings

Additional Fields

Drag fields to aggregate or group them.

Search

Add All Remove All

Grouped Fields

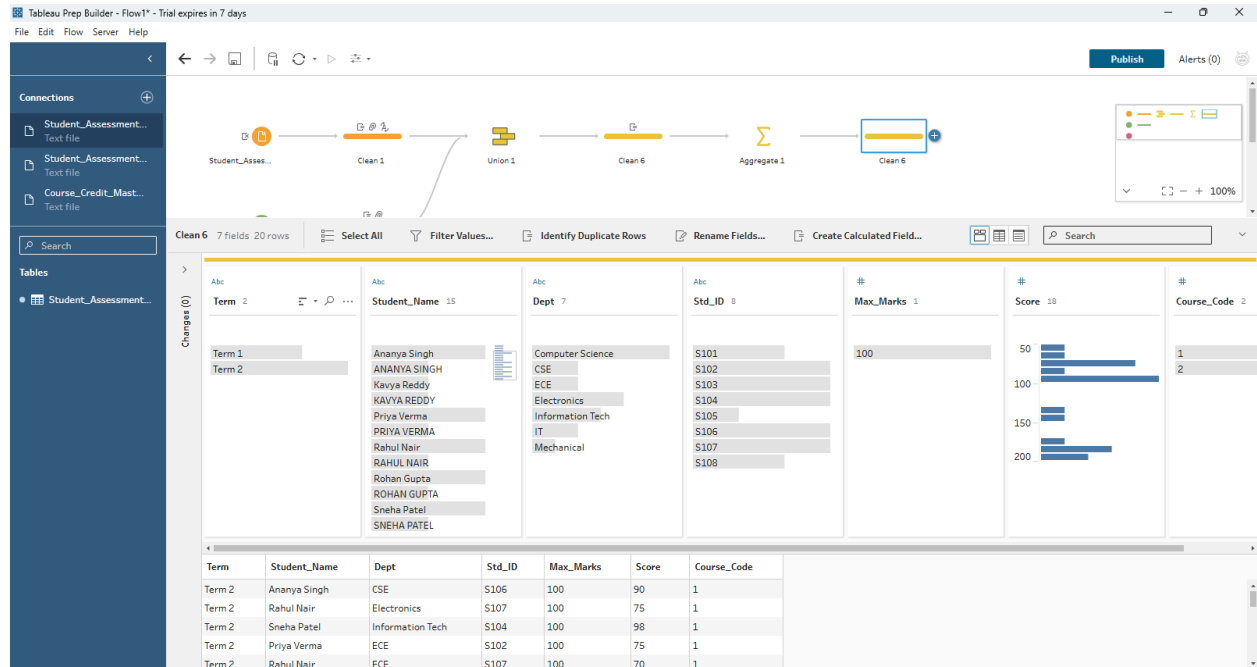
Drop fields here to group them

Aggregated Fields

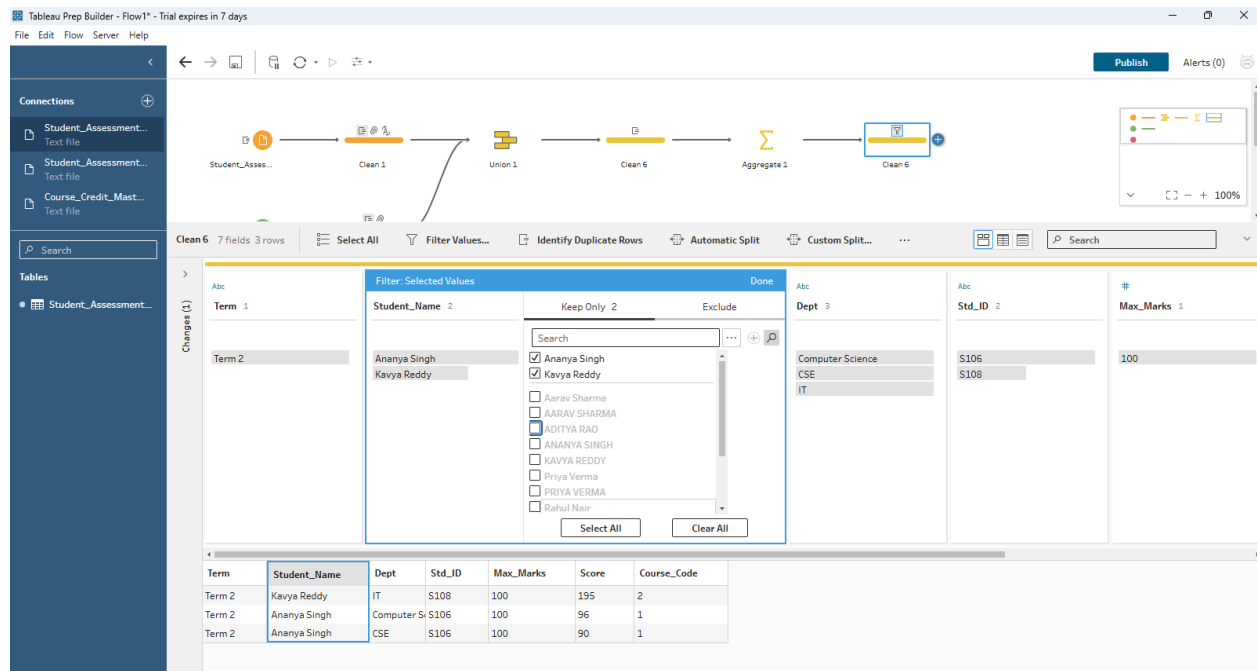
Drop fields here to aggregate them

GROUP	Term	Course_Code	Dept	Enrollment_Status	Exam_Date	Exam_Type	Max_Marks	Score	Std_ID	Student_Name	Table Names	Number of Rows (Aggregated)
GROUP	Term	Course_Code	Dept	Enrollment_Status	Exam_Date	Exam_Type	Max_Marks	Score	Std_ID	Student_Name	Table Names	Number of Rows (Aggregated)

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F: Apply Filters and Transformations



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File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Course_Credit_Mast... Text file
- Student_Assessment... Text file

Tables

- Student_Assessment...

10 fields 11 rows

Select All Filter Values... Identify Duplicate Rows

Search

100%

Letter_Grade 2

Term_Percentage 10

Term 2

Full_Name 11

Academic_Standing 1

Std_ID 8

D (Pass Class)

F (Fail / Re-appear)

18.75

27.5

36.25

39.25

44.25

45.5

46.5

47.5

48

48.75

39.25

1 (9%) row

Term 1

Term 2

AARAV SHARMA

ADITYA RAO

Ananya Singh

ANANYA SINGH

Kavya Reddy

KAVYA REDDY

Priya Verma

Rahul Nair

Rohan Gupta

Sneha Patel

SNEHA PATEL

Academic Probation (<60%)

S101

S102

S103

S104

S105

S106

S107

S108

Letter_Grade	Term_Percentage	Term	Full_Name	Academic_Standing	Std_ID	Dept	Score	Total_Possible
D (Pass Class)	48.75	Term 2	Kavya Reddy	Academic Probation (<60%)	S108	Information Tech	97.5	200

G: Create and Automate Data Preparation Flows

Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables

- Student_Assessment...

10 fields

Save output to

File

Browse

Name

Output

Location

C:\Users\ltab\Documents\My Tableau Prep Repository\Datasources

Output type

Tableau Data Extract (.hyper)

Write Options

Select an option to create or update your output table.

Full refresh

Create table

Run Flow

Save to Output.hyper

Letter_Grade	Term_Percentage	Academic_Standing	Term	Student_Name	Dept	Std_ID	Max_Marks	Score	Cour
null	null	null	Term 2	Ananya Singh	CSE	S106	100	90	1
null	null	null	Term 2	Kavya Reddy	IT	S108	100	195	2
null	null	null	Term 2	Ananya Singh	Computer Science	S106	100	96	1

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File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Course_Credit_Mast... Text file
- Student_Assessment... Text file

Tables

- Student_Assessment...

Save output to

File

Name

Output

Location

C:\Users\DELL\Documents\My Tableau Prep Repository\Datasources

Output type

Comma Separated Values (.csv)

Write Options

Select an option to create or update your output table.

Full refresh

Create table

Run Flow

Finished Running Flow

Output.csv

Total time 00:03

Done

Full_Name	Std_ID	Dept	Score	Tot
Kavya Reddy	S108	Information Tech	97.5	200
ADITYA RAO	S105	Mechanical	37.5	200
Priya Verma	S102	Electronics	78.5	200
Ananya Singh	S106	Computer Science	93	200
SNEHA PATEL	S104	Information Tech	93	200
Rahul Nair	S107	Electronics	72.5	200
ANANYA SINGH	S106	Computer Science	91	200
Sneha Patel	S104	Information Tech	96	200
KAVYA REDDY	S108	Information Tech	95	200
Rohan Gupta	S103	Computer Science	55	200
AARAV SHARMA	S101	Computer Science	88.5	200

D (Pass Class)	48	Term 2
D (Pass Class)	47.5	Term 1
F (Fail / Re-appear)	27.5	Term 2
D (Pass Class)	44.25	Term 1

```
C:\Users\DELL>"C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\tableau-prep-cli.bat" -t "C:\Users\DELL\Documents\My Tableau Prep Repository\Flows\Student_Performance_Prep_Flow.tfl"
JAVA_HOME is set to : C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\..\Plugins\jre temporarily
The connection file is in V1 format.
Preparing to run the flow : C:\Users\DELL\Documents\My Tableau Prep Repository\Flows\Student_Performance_Prep_Flow.tfl
Loading the flow.
Creating temp directory at C:\Users\DELL\AppData\Local\Temp\prep-cli-2834400816672830943
Product license verified.
Loaded the flow.
Checking the flow document for errors.
Flow Document has no errors.
Preparing to execute the flow.
Flow Execution Status: Running
Flow Execution Status: Finished
Finished running the flow successfully.
C:\Users\DELL>
```

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